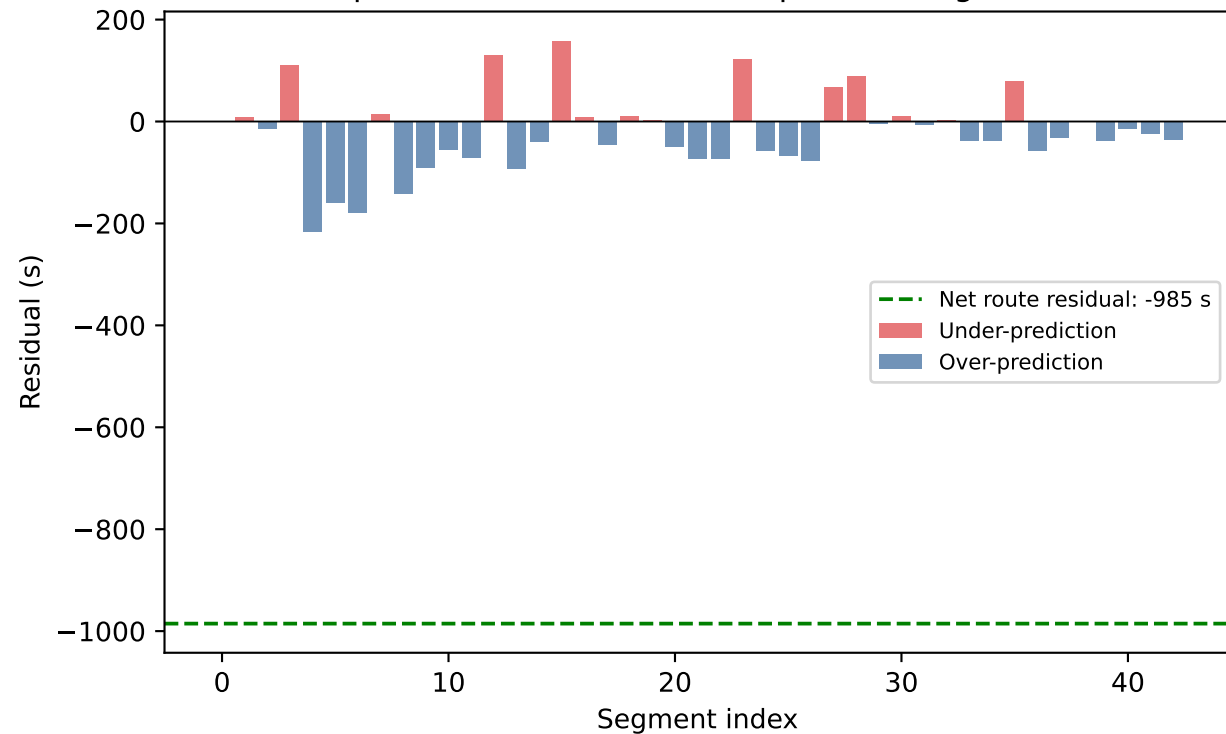


Panel A -- Per-segment residuals
(representative calibration trip, n=43 segments)



Panel B -- Route residual distributions (W4 calibration)
30.5% variance reduction from residual diversification

